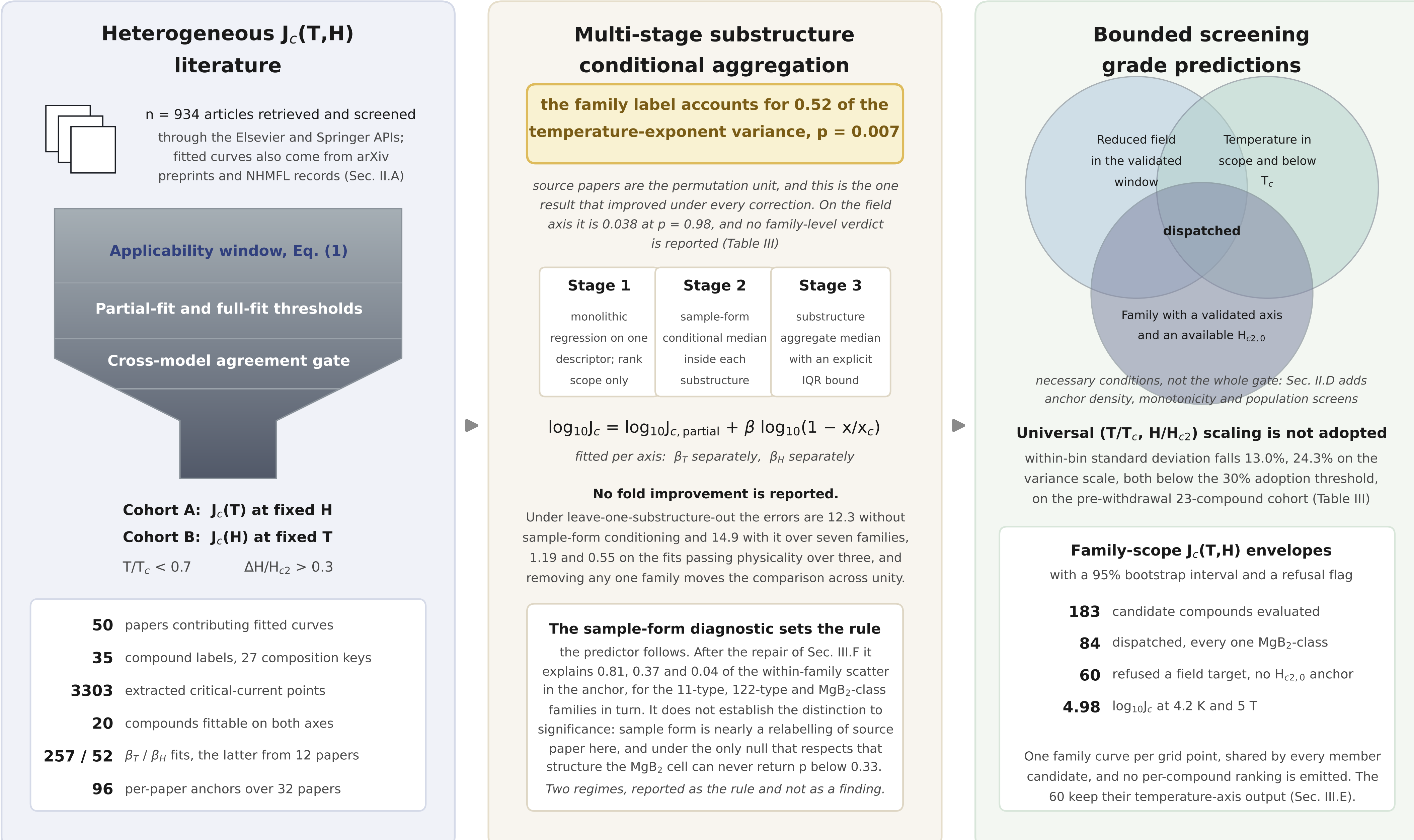
Heterogeneous $J_c(T,H)$ literature
n = 934 articles retrieved and screened
through the Elsevier and Springer APIs; fitted curves also come from arXiv preprints and NHMFL records (Sec. II.A)
Applicability window, Eq. (1)
Partial-fit and full-fit thresholds
Cross-model agreement gate
Cohort A: $J_c(T)$ at fixed H
Cohort B: $J_c(H)$ at fixed T
$T/T_c < 0.7$
$\Delta H/H_{c2} > 0.3$
50 papers contributing fitted curves
35 compound labels, 27 composition keys
3303 extracted critical-current points
20 compounds fittable on both axes
257 / 52 β_T / β_H fits, the latter from 12 papers
96 per-paper anchors over 32 papers
Multi-stage substructure conditional aggregation
the family label accounts for 0.52 of the temperature-exponent variance, p = 0.007
source papers are the permutation unit, and this is the one result that improved under every correction. On the field axis it is 0.038 at p = 0.98, and no family-level verdict is reported (Table III)
Stage 1
monolithic regression on one descriptor; rank scope only
Stage 2
sample-form conditional median inside each substructure
Stage 3
substructure aggregate median with an explicit IQR bound
$\log_{10} J_c = \log_{10} J_{c,\,\mathrm{partial}} + \beta \log_{10}(1 - x/x_c)$
fitted per axis: β_T separately, β_H separately
No fold improvement is reported.
Under leave-one-substructure-out the errors are 12.3 without sample-form conditioning and 14.9 with it over seven families, 1.19 and 0.55 on the fits passing physicality over three, and removing any one family moves the comparison across unity.
The sample-form diagnostic sets the rule
the predictor follows. After the repair of Sec. III.F it explains 0.81, 0.37 and 0.04 of the within-family scatter in the anchor, for the 11-type, 122-type and MgB_2-class families in turn. It does not establish the distinction to significance: sample form is nearly a relabelling of source paper here, and under the only null that respects that structure the MgB_2 cell can never return p below 0.33.
Two regimes, reported as the rule and not as a finding.
Bounded screening grade predictions
Reduced field in the validated window
Temperature in scope and below T_c
dispatched
Family with a validated axis and an available $H_{c2,0}$
necessary conditions, not the whole gate: Sec. II.D adds anchor density, monotonicity and population screens
Universal $(T/T_c,\ H/H_{c2})$ scaling is not adopted
within-bin standard deviation falls 13.0%, 24.3% on the variance scale, both below the 30% adoption threshold, on the pre-withdrawal 23-compound cohort (Table III)
Family-scope $J_c(T,H)$ envelopes
with a 95% bootstrap interval and a refusal flag
183 candidate compounds evaluated
84 dispatched, every one MgB_2-class
60 refused a field target, no $H_{c2,0}$ anchor
4.98 $\log_{10} J_c$ at 4.2 K and 5 T
One family curve per grid point, shared by every member candidate, and no per-compound ranking is emitted. The 60 keep their temperature-axis output (Sec. III.E).